

Functional sequences, series and spaces – 5

Hilbert spaces and orthogonal systems

13 сентября 2026 года

Theoretical minimum.

Definition. A Hilbert space is a complete space with an inner product; the model examples are ℓ^2 and $L^2[a, b]$ with $\langle f, g \rangle = \int f \bar{g}$. A system (e_n) is orthonormal if $\langle e_n, e_m \rangle = \delta_{nm}$, and complete if its closed linear span is the whole space.

Theorem 1 (projection and Bessel). The best approximation to x by elements of a closed subspace L is the orthogonal projection, characterised by $x - Px \perp L$; for an orthonormal system the coefficients of the best approximation are $c_n = \langle x, e_n \rangle$ and $\sum |c_n|^2 \leq \|x\|^2$ (Bessel). Equality for every x — Parseval's identity — holds if and only if the system is complete.

Theorem 2 (Riesz–Fischer). For an orthonormal system and any $(c_n) \in \ell^2$ the series $\sum c_n e_n$ converges in the space; consequently a separable infinite-dimensional Hilbert space is isometric to ℓ^2 , and the Gram–Schmidt process turns any dense independent system into an orthonormal basis.

Fact 1 (orthogonal polynomials). Applying Gram–Schmidt to $1, x, x^2, \dots$ with a weight w gives a system satisfying a three-term recurrence; its n -th member has n simple roots inside the interval, and consecutive members interlace. The weights 1 , $(1 - x^2)^{-1/2}$ and e^{-x^2} give the Legendre, Chebyshev and Hermite polynomials.

Fact 2 (weak convergence). An orthonormal sequence tends to zero weakly but not in norm, so the unit ball is not compact; this is the standard source of counterexamples in a Hilbert space.

1. Prove Bessel's inequality, the projection theorem in the form " $x - Px \perp L$ characterises the best approximation", and the equivalence "Parseval for every $x \Leftrightarrow$ the system is complete".
2. Prove that the trigonometric system $\frac{1}{\sqrt{2}}, \cos x, \sin x, \cos 2x, \sin 2x, \dots$ is complete in $L^2[0, 2\pi]$ — deduce it from the density of trigonometric polynomials (Continuous functions – 9) — and write down what Parseval's identity says for a continuous 2π -periodic function. Then prove that a continuous function orthogonal to all e^{inx} , $n \in \mathbb{Z}$, vanishes identically. (The problem about an equidistant set in an infinite-dimensional Hilbert space is in Linear algebra – 7.)
3. (Kac 1.90) Let $(n_k)_{k \geq 0}$ be an increasing sequence of integers with $n_0 = 0$. Prove that the system of functions $(x^{n_k})_{k \geq 0}$ is complete in $C[0, 1]$ if and only if $\sum_{k \geq 1} \frac{1}{n_k}$ diverges. (Start with completeness in the L^2 norm; the statement itself was quoted without proof in Continuous functions – 9.)
4. (Kac 1.88) Let $U: H \rightarrow H$ be a unitary operator on a Hilbert space. Prove that for every $x \in H$ the averages $\frac{1}{n} \sum_{i=1}^n U^i x$ converge to Px , where P is the orthogonal projection onto the subspace of U -invariant vectors. Deduce (Kac 1.86) that for a two-sided sequence $c \in \ell^2(\mathbb{Z})$ the shifts satisfy $\left\| \frac{T_0 c + T_1 c + \dots + T_{n-1} c}{n} \right\| \rightarrow 0$.
5. Construct the Legendre polynomials by applying Gram–Schmidt to $1, x, x^2, \dots$ on $[-1, 1]$, prove the three-term recurrence, and prove that the roots are simple, lie in $(-1, 1)$ and interlace for consecutive degrees.